

Bayesian Networks vs. Machine Learning for Postnatal Depression Prediction: An Interpretability-Robustness Trade-off

Tisha Shah¹, Pooja Vartak¹, and Kanchan Dabre¹

Department of Computer Science and Engineering (Data Science),
Dwarkadas J. Sanghvi College of Engineering, Mumbai, India
{shahtisha16@gmail.com,pooja.vartak@djsce.ac.in,kanchan.dabre@djsce.ac.in}

Abstract. Prediction of anxiety in screening for postnatal depression is complicated by uncertain symptom-outcome relationships and often incomplete questionnaire data. This paper compares four Bayesian network structure-learning methods (Chow-Liu tree search, hill-climbing search, the PC algorithm and naive Bayes) against three machine learning classifiers (logistic regression, random forest and XGBoost) using a postnatal depression dataset of 1,491 respondents. The seven models are compared in two versions of the dataset (real missing values preserved, and fully complete) under a leak-free train-test protocol, and in a controlled experiment that injects missingness into the test set (10–40%) in fixed, already-trained models to separate inference-time robustness from structure-learning requirements. The PC algorithm obtains the highest accuracy (up to 97.12%) but the densest, least interpretable structure, thus confirming a quantifiable complexity-accuracy trade-off. All four variants of the Bayesian network degrade more gracefully with increasing missingness than the machine learning classifiers, with native missing-value handling in XGBoost not preventing a sharp drop in accuracy (89.91% to 61.67% at 40% missingness) as often assumed. This pattern is similar for both variants of the dataset. Bayesian networks thus offer a robustness advantage for incomplete clinical data that machine learning models assumed to handle missingness natively do not automatically match.

Keywords: postnatal depression · Bayesian networks · structure learning · missing data · machine learning · XGBoost · interpretability

1 Introduction

1.1 Problem Statement

Postnatal depression is a mood disorder affecting mothers during the early postpartum period, frequently accompanied by related symptoms such as persistent sadness, irritability, guilt, disrupted sleep, and, in severe cases, suicidal ideation. This paper considers the prediction of the anxiety, a closely related and clinically relevant outcome, from a structured questionnaire recording demographic

information and self-reported symptoms from a mother . You are given a set of categorical features (age group, sadness, irritability, sleep disturbance, concentration difficulty, appetite change, guilt, bonding difficulty, history of suicide attempt) and the task is to predict whether a respondent reports feeling anxious or not. This is a binary classification problem, and there are two aspects that make it different from a typical tabular task: the relationship between symptoms and outcome is not deterministic, but uncertain, and the input data, collected through a real clinical questionnaire, is often incomplete, as respondents may refuse to answer sensitive items such as those related to suicide attempts.

Identifying mothers at risk early enables timely clinical intervention to reduce severity and duration of the condition and downstream effects on infant development. Ideally, a model to support such screening should be *interpretable* so that a clinician can understand which factors drive a given risk estimate, and *robust to missing data* as real questionnaire responses are rarely fully complete. The same model is not always able to satisfy both requirements simultaneously. Understanding this trade-off for clinical screening data is the central motivation for this study.

Random forest or gradient-boosted variants generally performed best in previous research on predicting postpartum depression, which looked at combinations of support vector machines, random forests, logistic regression, naive Bayes, gradient boosting, and neural networks [2, 8, 7]. Some recent works emphasize interpretability with post hoc methods such as SHAP [5, 6]. Furthermore, recent scoping reviews note that although Bayesian networks are theoretically appropriate for this reasoning task, they are much less commonly used than random forests or logistic regression in this literature, and that missing data is rarely systematically modeled, with incomplete records typically excluded entirely [3, 1, 4].

Structure-learning algorithms such as Chow-Liu tree construction [11], greedy hill-climbing search [12] and the constraint-based PC algorithm [13] have been compared independently in the Bayesian network literature on synthetic or benchmark data [14], but rarely on real clinical data and rarely with a quantified trade-off between structural complexity and accuracy . Bayesian network inference is known to still operate directly on incomplete evidence via marginalization separately [17, 18], and Bayesian networks have been argued to handle missing electronic health record data more naturally than other model families [19]. Only rarely has the property been tested against the native missing-value handling built into gradient-boosted trees such as XGBoost [20], which is often assumed to provide comparable robustness.

In brief, the literature treats the approaches based on Bayesian networks and machine learning as two distinct traditions for this problem, and no study identified in our review compares directly both families using the same postnatal depression dataset under a controlled, missing-data robustness protocol.

This paper addresses this gap. We make the following contributions.

1. We compare four variants of Bayesian network, learnt with different structure-learning algorithms (Chow-Liu tree search, hill-climbing search, the PC al-

gorithm, and a fixed naive Bayes structure), with three machine learning classifiers (logistic regression, random forest, XGBoost), in a single, identical, leak-free train-test protocol – to the best of our knowledge the first such joint comparison on this problem.

2. We propose an experiment for robustness to missing data, using a fixed pre-trained model of each type on data with increasing artificial missingness (10–40%, MCAR), to explicitly separate *inference*-time robustness from *structure-learning*-time robustness, which was not explicitly done in previous comparative work.
3. Instead of only describing the trade-off qualitatively, we quantify the trade-off between the complexity of the learned network and its accuracy.
4. We empirically show that the native missing-value handling of XGBoost does not provide the robustness that is commonly assumed, degrading substantially more than any of the four Bayesian network variants under identical missingness.
5. We conduct this evaluation on two versions of the dataset – one with true, naturally occurring missingness, and one that is fully complete – verifying the relative ordering of model robustness on both.

2 Related Work

Existing works on PPD prediction are limited to conventional machine learning classifiers, while post hoc interpretability has gained more attention. Table 1 summarizes the most important studies since 2021.

Table 1. Recent (2021 onward) studies on ML-based postpartum depression prediction.

Study	Year	Models	Gap relative to this work
Alkhateeb et al. [1]	2026	Review, 65 studies	Missing-data handling inconsistent/unreported.
Malani et al. [2]	2026	GBT, elastic net	No PGM comparison; n=205,495.
Xia et al. [3]	2025	Review	Notes BNs rarely used vs. RF/SVM/LR.
Wang et al. [4]	2025	Random forest	Missing EHR records excluded, not modeled.
Zhang et al. [5]	2025	XGBoost + SHAP	Interpretability only post hoc.
Huang et al. [6]	2025	LASSO + ensemble	No PGM or missingness analysis.
Qi et al. [7]	2025	LR, ANN, others	Missing data not a focus.
Kim [8]	2025	DT, RF, AdaBoost, LR	No PGM included.
Lin and Zhou [9]	2025	Ensemble NN	Trade-off resolved via ensembling, not PGMs.
JMIR [10]	2021	14-study review	No structure-learning comparison.

Three gaps are common across these studies: Bayesian networks are absent or limited to a single fixed structure; missing data is treated by exclusion or

single-pass imputation and not investigated directly; and interpretability, when addressed, is an add-on rather than a direct comparison to an inherently interpretable model family. This paper addresses all three together.

For Bayesian networks, there are several families of structure-learning algorithms, including Chow-Liu tree construction [11], greedy hill-climbing search [12], and constraint-based algorithms such as the PC algorithm [13]. Such algorithms were compared experimentally in [14] and it was found that hybrid methods like Max-Min Hill-Climbing [15] often outperform purely score-based or constraint-based search on benchmark networks. The comparisons are performed on synthetic or curated data, and none of them reports the trade-off between structural complexity and accuracy on a small real clinical dataset.

Bayesian network inference can directly handle incomplete evidence by marginalising over unobserved variables [18], extending the EM approach of Dempster et al. [16]. Lin and Haug [17] compared naive Bayes and structure-learned networks with several missing-data treatments and McGeachie et al. [19] argue that Bayesian networks can handle missing EHR data more naturally than alternatives such as SVMs. Chen and Guestrin [20] describe the native sparsity-aware treatment of missing values by XGBoost, which is often assumed to yield the robustness of probabilistic inference – an assumption tested directly in Section 5.

3 Methodology

3.1 Problem Formulation

Let $X = (X_1, \dots, X_9)$ denote the nine categorical feature variables (Age, Feeling sad or Tearful, Irritable towards baby & partner, Trouble sleeping at night, Problems concentrating, Overeating or loss of appetite, Feeling of guilt, Problems of bonding with baby, Suicide attempt) and let $Y \in \{0, 1\}$ denote the target variable, *Feeling anxious*. The task is to estimate $P(Y \mid X = x)$ for a given observed feature vector x , and to produce a class prediction $\hat{y} = \arg \max_y P(Y = y \mid X = x)$. Critically, in deployment x may be only partially observed; a subset of features may be missing for a given respondent, and the model must still produce an estimate of $P(Y \mid X_{\text{obs}})$ using only the observed subset $X_{\text{obs}} \subseteq X$.

Figure 1 summarizes the overall methodology followed in this study, from raw data preprocessing through the construction of two baseline dataset variants, model training, and the two evaluation tracks (baseline-only evaluation and the missing-data robustness experiment).

We address this task using two model families: Bayesian networks, in which the joint distribution $P(X, Y)$ is factorized according to a learned directed acyclic graph (DAG) structure, and conventional machine learning classifiers, which directly model $P(Y \mid X)$ without an explicit graphical structure.

3.2 Bayesian Network Structure Learning

A Bayesian network represents $P(X, Y)$ as a product of conditional distributions, $P(X, Y) = \prod_i P(Z_i \mid \text{Pa}(Z_i))$, where Z_i ranges over all ten variables and $\text{Pa}(Z_i)$

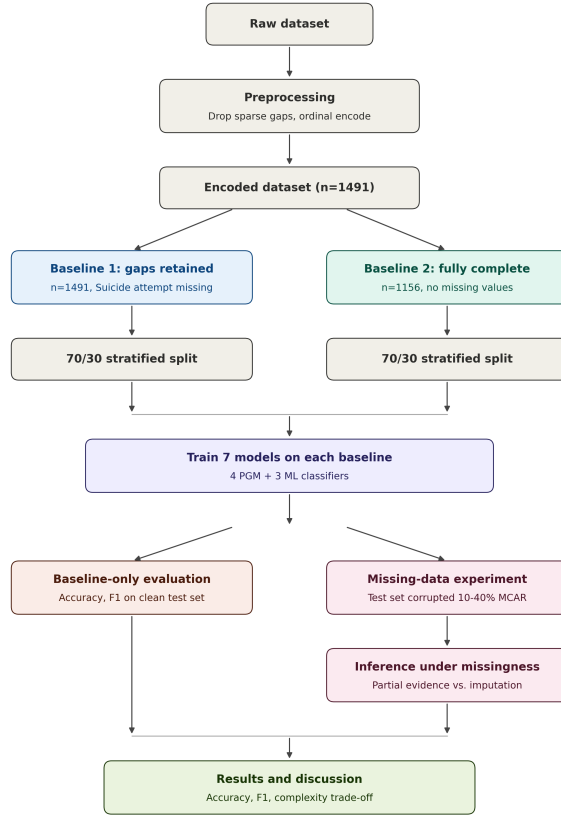


Fig. 1. Overall methodology pipeline followed in this study.

denotes the parents of Z_i in the DAG. Because the true DAG is not known a priori, we learn it using four distinct structure-learning algorithms.

Chow-Liu tree search. This method restricts each variable to at most one parent and finds the tree that maximizes data likelihood by computing the pairwise mutual information

$$I(X_i, X_j) = \sum_{x_i, x_j} \hat{P}(x_i, x_j) \log \frac{\hat{P}(x_i, x_j)}{\hat{P}(x_i)\hat{P}(x_j)} \quad (1)$$

for every variable pair, then extracting the maximum-weight spanning tree over the resulting fully connected graph [11].

Hill-climbing search. This method performs an unrestricted greedy local search over the space of DAGs, at each step adding, removing, or reversing a

Algorithm 1 Chow-Liu Tree Search

- 1: Compute $I(X_i, X_j)$ for every pair of variables
 - 2: Build a complete weighted graph with edge weights $I(X_i, X_j)$
 - 3: Extract the maximum-weight spanning tree
 - 4: Orient edges outward from a chosen root node
 - 5: Estimate $P(Z_i \mid \text{Pa}(Z_i))$ for each node via MLE
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single edge to improve the Bayesian Information Criterion (BIC),

$$\text{BIC}(G) = \log P(D \mid \hat{\theta}, G) - \frac{|\theta|}{2} \log N, \quad (2)$$

where D is the data, $\hat{\theta}$ the maximum-likelihood parameters under structure G , $|\theta|$ the number of free parameters, and N the number of training rows [12]. The penalty term discourages unnecessarily dense structures.

Algorithm 2 Hill-Climbing Search

- 1: Initialize G as an empty graph
 - 2: **repeat**
 - 3: Consider all single-edge additions, removals, reversals to G that keep G acyclic
 - 4: Select the operation that most improves $\text{BIC}(G)$
 - 5: Apply it if it improves the score; otherwise stop
 - 6: **until** no operation improves the score
-

PC algorithm. This constraint-based method recovers the network structure using conditional independence tests rather than a scoring function. We use the chi-square test of independence between each variable pair, conditioned on increasing-size subsets of their neighbors, removing an edge whenever a conditioning set renders the pair independent [13].

Algorithm 3 PC Algorithm (Skeleton + Orientation)

- 1: Start from a complete undirected graph over all variables
 - 2: **for** increasing conditioning set size $k = 0, 1, 2, \dots$ **do**
 - 3: **for** each adjacent pair (X_i, X_j) **do**
 - 4: **if** $X_i \perp X_j \mid S$ for some S of size k among their neighbors **then**
 - 5: Remove edge (X_i, X_j)
 - 6: **end if**
 - 7: **end for**
 - 8: **end for**
 - 9: Orient remaining edges using v-structure and acyclicity rules
-

Naive Bayes. This method fixes the structure in advance rather than learning it, placing the target Y as the sole parent of every feature and assuming all features are conditionally independent given Y : $P(X, Y) = P(Y) \prod_i P(X_i | Y)$.

Parameter estimation. For all four structures, the conditional probability tables are estimated by Maximum Likelihood Estimation (MLE), $\hat{P}(Z_i = z | \text{Pa}(Z_i) = p) = \frac{\text{count}(Z_i=z, \text{Pa}(Z_i)=p)}{\text{count}(\text{Pa}(Z_i)=p)}$, using only fully observed training rows, since none of the four structure-learning algorithms used here can directly process incomplete training data.

Inference. We perform exact inference using variable elimination, which computes $P(Y | X_{\text{obs}})$ by summing out every variable not in X_{obs} or Y . Because variable elimination only conditions on the evidence actually supplied, it can be queried with an arbitrary subset of observed features, which is the property exploited in our missing-data robustness experiment (Section 4).

3.3 Machine Learning Classifiers

Logistic regression models $P(Y = 1 | x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$, with w, b fit by minimizing log loss.

Random forest [?] averages predictions across T independently bootstrap-trained decision trees, $\hat{P}(Y = 1 | x) = \frac{1}{T} \sum_{t=1}^T 1[h_t(x) = 1]$. We constrain tree depth (max depth 5, minimum 5 samples per leaf) after observing severe overfitting (100% training accuracy) with unconstrained trees.

XGBoost [20] builds an additive ensemble of regression trees, $\hat{y}_i = \sum_{k=1}^K f_k(x_i)$, fit by minimizing a regularized objective $\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$, where Ω penalizes tree complexity. XGBoost natively routes missing feature values during training and inference via a learned default split direction, requiring no imputation. We similarly constrain depth (max depth 3, minimum child weight 15) after observing a large train-test accuracy gap under default settings.

3.4 Missing-Data Handling Strategy

Table 2 summarizes how each model type handles missing values, reflecting how each is actually used in practice rather than applying one uniform imputation rule to all seven models.

Table 2. Missing-data handling strategy by model type.

Model	Missing-data strategy
Tree Search, Hill Climb, PC, Naive Bayes	Variable elimination with partial evidence at inference time; complete rows only for structure/parameter learning
Logistic Regression, Random Forest	Mode imputation prior to training and inference
XGBoost	Native sparsity-aware split routing; no imputation

3.5 Evaluation Metrics

Given the moderate class imbalance in the target variable (Section ??), we report both accuracy, $\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$, and the F1-score, $F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$, for every model and experimental condition.

4 Experiments

4.1 Dataset and Baseline Variants

We used a dataset of 1,503 responses to a postnatal depression questionnaire collected at a medical hospital, consisting of nine categorical features and a binary target variable, *Feeling anxious*. After dropping the non-predictive timestamp column and 27 rows with sparse, unstructured missing values in three minor fields, we had 1,491 rows remaining. A further 335 of these rows (22.5%) have a missing value in the *Suicide attempt* field, originally coded as “Not interested to say” rather than a blank entry; we treat this as a missing value rather than a third ordinal category, since non-disclosure of a suicide attempt carries no information about the underlying answer.

We evaluate all seven models on two variants of the dataset, as detailed in Section 3. **Baseline 1** keeps the 1,491 row dataset, and the natural missingness in the *Suicide attempt* field is also kept as it is, which would have been the case in realistic questionnaire conditions. **Baseline 2** takes it one step further and removes the 335 rows which have missing *Suicide attempt* value, so there is a full 1,156-row dataset. Instead of an arbitrary label encoding, we encode all categorical features with a manual ordinal encoding (e.g., No $\rightarrow$ 0, Sometimes $\rightarrow$ 1, Yes $\rightarrow$ 2), so as to preserve the natural severity ordering in each variable.

Each baseline is independently split into training and test sets using a stratified 70/30 split (fixed random seed, stratified on *Feeling anxious*) resulting in 1,043 training and 448 test rows for Baseline 1 and 809 training and 347 test rows for Baseline 2. Every model is trained and evaluated on exactly the same rows, as the same split is reused over all seven models within each baseline in exactly the same way.

4.2 Baseline-Only Evaluation Protocol

For each baseline, we train all seven models – four Bayesian network variants and three machine learning classifiers – on the training split, and evaluate them once on the held-out test split, using the missing-data handling strategy described in Table 2. We have developed this protocol to address a methodological problem we faced in previous publications that employed a similar data set and modeling approach: model accuracy was evaluated on the same data used for training. In such a case, the accuracy reported for a structure-learned Bayesian network and an unconstrained Hill-Climbing network was about 38 percentage points apart, a difference that did not reappear when a proper held-out test set was used. All reported accuracy and F1-score values in Section 5 are computed only on held-out test data unseen during training.

4.3 Missing-Data Robustness Experiment Design

The central experiment of this study evaluates how each model’s predictive performance degrades as the proportion of missing values in the input data increases. A naive design would inject missingness into both the training and test data and relearn each model from scratch at every missingness level; we initially attempted this and found that it conflates two distinct effects: the algorithm’s ability to learn a structure or fit parameters from incomplete training data, and the algorithm’s ability to produce a prediction from incomplete test-time evidence. Because Bayesian network structure-learning algorithms require complete rows to compute the statistics they rely on (mutual information for Chow-Liu, BIC scores for Hill-Climbing, conditional independence tests for the PC algorithm), injecting missingness into the training data caused the number of usable training rows to collapse sharply – from 816 complete rows at 0% injected missingness to only 8 complete rows at 40% injected missingness on Baseline 2’s training split – making the resulting structures and their accuracy unreliable for reasons unrelated to the models’ actual robustness.

We therefore separate these two effects explicitly. Each of the seven models is trained exactly once, on the unmodified training split of a given baseline, exactly as in the baseline-only evaluation. The resulting trained model – a fixed Bayesian network structure with its conditional probability tables, or a fixed fitted classifier – is then evaluated repeatedly against test sets with increasing artificial missingness, without any retraining. This design isolates the robustness of each model’s *inference* procedure under missing evidence from the separate question of whether its *structure-learning* procedure could have tolerated the same missingness during training, a distinction that is not made explicit in prior comparative studies of this kind.

Artificial missingness is injected following a missing-completely-at-random (MCAR) mechanism: for a target missingness rate $p \in \{0.10, 0.20, 0.30, 0.40\}$ and each feature column independently, a uniformly random subset of rows of size $p \times n$ is selected (without replacement) and the corresponding cell is set to missing. This is applied independently to every feature column, using a fixed random seed per missingness level for reproducibility. Because *Suicide attempt* already carries real missingness from Baseline 1 prior to this injection, the realized overall missingness rate is slightly higher than the nominal target in that column; Table 3 reports both the nominal and realized rates.

Table 3. Nominal versus realized missingness rates after MCAR injection (Baseline 1 training split).

Nominal rate	Realized rate (all feature cells)
10%	12.24%
20%	22.00%
30%	31.55%
40%	41.53%

At each missingness level, the four Bayesian network models are queried using variable elimination with whatever subset of features remains observed for a given test row, exploiting the partial-evidence inference property described in Section 3. Logistic regression and random forest, which cannot accept missing inputs directly, have the missing cells filled in using the mode of each column computed from the (unmodified) training data, mirroring how these models are used in practice. XGBoost is given the corrupted test data directly, relying on its native sparsity-aware handling of missing values without any imputation step. This experiment is run independently on both Baseline 1 and Baseline 2, allowing us to confirm in Section 5 whether the resulting pattern of relative model robustness depends on which baseline is used as the starting point.

5 Results and Discussion

5.1 Baseline-Only Performance

Table 4 summarizes test-set accuracy for all seven models on Baseline 1 (real-world gaps retained), Baseline 2 (fully complete data), and after 40% artificial test-set missingness starting from Baseline 2. The PC algorithm achieves the highest accuracy on both baselines (95.88%, 97.12%), followed by XGBoost and random forest; Tree Search and logistic regression are weakest. Every model improves from Baseline 1 to Baseline 2 except logistic regression (83.04% to 80.98%), likely because Baseline 2 is not a random subsample of Baseline 1 but specifically the respondents who answered *Suicide attempt*, whose target class balance is more skewed (72.66%/27.34% versus 64.92%/35.08%).

Table 4. Summary of all seven models across both baselines and the missing-data robustness test (40% test-set missingness, Baseline 2).

Model	Type	Edges	Base. 1 Acc	Base. 2 Acc	Acc @ 40% miss.
PC Algorithm	PGM	39–40	95.88%	97.12%	89.63%
Hill Climb	PGM	21	85.00%	87.03%	81.27%
Naive Bayes	PGM	9	80.00%	83.57%	79.83%
Tree Search	PGM	9	77.35%	80.40%	76.08%
Random Forest	ML	–	86.83%	91.93%	79.83%
XGBoost	ML	–	88.84%	89.91%	61.67%
Logistic Regr.	ML	–	83.04%	80.98%	76.66%

Figure 2 visualizes this comparison directly; the consistent improvement on Baseline 2 and the gap between PGM and ML groups are both visible at a glance.

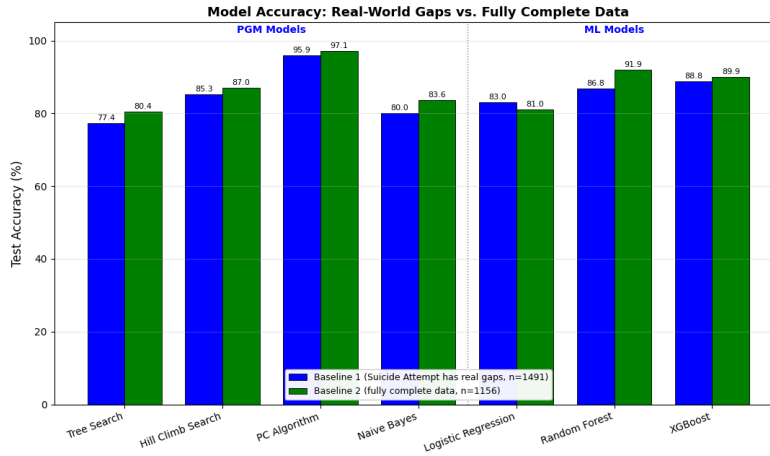


Fig. 2. Test accuracy on Baseline 1 versus Baseline 2 for all seven models.

5.2 Structural Complexity and the Interpretability-Accuracy Trade-off

The four Bayesian network variants range from 9 edges (Tree Search, Naive Bayes) to approximately 39–40 edges (PC algorithm; this count varies marginally between runs due to the algorithm’s internal tie-breaking, though the resulting accuracy is stable). Figure 3 plots complexity against Baseline 2 test accuracy, showing a clear monotonic relationship: each increase in structural complexity brings a further accuracy gain.

Figure 4 shows the PC algorithm’s structure: 40 edges among 10 variables, most connected to several others, versus the 9-edge Tree Search chain where each variable has at most one parent and the path to *Feeling anxious* is easy to trace. The PC algorithm’s denser structure gives the highest accuracy here but is far harder to interpret – the central trade-off this paper documents, consistent with findings elsewhere in clinical ML [?,21].

The PC algorithm’s result also warrants caution: approximately 2.6% of its test predictions produced an undefined (0/0) probability, due to evidence combinations unseen during training – a direct consequence of its structure being dense relative to the size of the training data (809 rows). We report its accuracy as observed but do not treat it as an unqualified best result.

5.3 Missing-Data Robustness

Figure 5 shows test accuracy versus injected test-set missingness (0–40%, MCAR) for models trained once on Baseline 2. The four Bayesian network variants degrade gradually (Naive Bayes: −3.75 points; Tree Search: −4.32; Hill-Climbing: −5.76; PC algorithm: −7.49, despite starting highest). The three ML classifiers degrade considerably more (logistic regression: −4.32; random forest: −12.10;

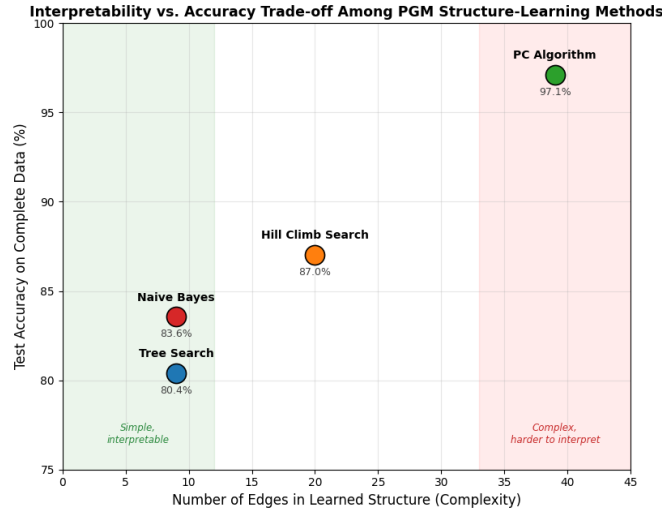


Fig. 3. Test accuracy versus number of edges in the learned Bayesian network structure (Baseline 2).

XGBoost: -28.24 , falling from 89.91% to 61.67% , near majority-class level), despite XGBoost natively handling missing values without imputation.

To our knowledge, this is a novel observation for this application: XGBoost’s native handling of missing values does not confer robustness comparable to Bayesian network inference. Variable elimination computes $P(Y \mid X_{\text{obs}})$ by exactly marginalizing over whatever is unobserved, using the full learned joint distribution. XGBoost instead learns a single default split direction per tree node from missingness patterns seen *during training*; as test-time missingness grows well beyond what was seen in training, this learned default becomes a progressively weaker substitute for the real value.

Repeating this experiment from Baseline 1 reproduces the same ordering at lower absolute accuracy (PC algorithm: $86.16\% \rightarrow 82.59\%$, -3.57 ; XGBoost: $88.84\% \rightarrow 70.09\%$, -18.75), confirming the finding does not depend on starting from fully complete data.

5.4 Additional Observation: Age as a Weak Predictor

The Hill-Climbing search excluded *Age* from its structure entirely, since no edge involving it improved the BIC score enough to justify inclusion. This agrees with feature importance scores from both tree-based ML models: *Age* ranked lowest of all nine features in random forest (2.72%) and XGBoost (2.89%). Three independent methods – one structural, two importance-based – agree that age contributes little predictive signal once symptom-level variables are known.

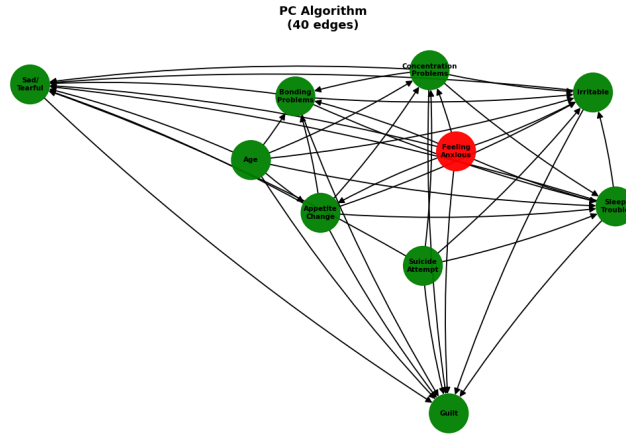


Fig. 4. Bayesian network structure learned by the PC algorithm (one representative run; edge count varies marginally between runs due to internal tie-breaking).

5.5 Limitations and Future Work

This study has several limitations. The injected missingness follows an MCAR mechanism; real non-response to sensitive items such as *Suicide attempt* is more plausibly MAR or MNAR, and future work should repeat this experiment under those mechanisms. The four structure-learning algorithms required complete training rows, so their ability to learn structure (as opposed to perform inference) was not itself stress-tested under missing training data – an inherent limitation of these algorithms, not of our design. This study uses a single hospital dataset, and the relative robustness ordering observed here should be validated on independent datasets before being treated as general. Baseline 2’s respondents differ systematically from Baseline 1’s in class balance, limiting direct cross-baseline comparison of absolute accuracy, though relative model ordering is preserved in both. Finally, the PC algorithm’s high accuracy comes with a dense structure, limited interpretability, and a small but non-zero rate of undefined predictions, and should be weighed against the use case rather than read as an unqualified recommendation. Future work could extend this comparison to methods such as Max-Min Hill-Climbing [15], evaluate calibration, and validate learned structures against clinical expertise.

6 Conclusion

This paper presented a controlled comparison of four Bayesian network structure-learning methods (Chow-Liu tree search, hill-climbing search, the PC algorithm, and naive Bayes) against three machine learning classifiers (logistic regression, random forest, and XGBoost) for predicting anxiety from a real postnatal depression questionnaire dataset. Under a leak-free evaluation protocol, the PC

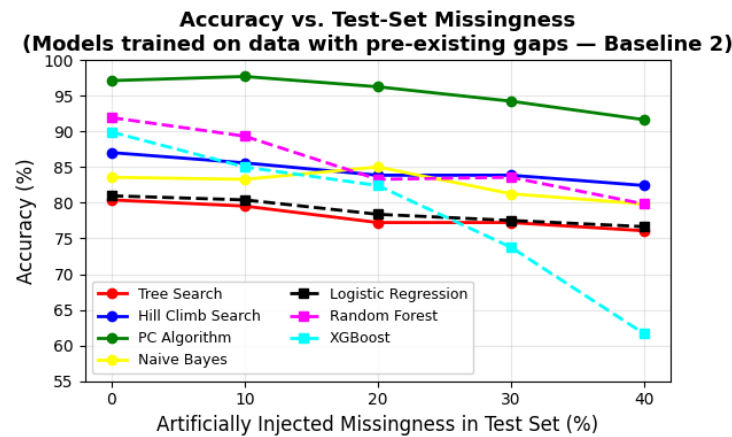


Fig. 5. Test accuracy versus injected test-set missingness, models trained on Baseline 2.

algorithm achieved the highest accuracy but produced the densest, least interpretable structure, confirming a quantifiable trade-off between structural complexity and predictive accuracy among Bayesian network variants. A controlled missing-data robustness experiment, designed to isolate inference-time robustness from structure-learning requirements, showed that all four Bayesian network variants degraded considerably more gracefully under increasing test-set missingness than the machine learning classifiers, including XGBoost, whose native missing-value handling did not prevent a substantial drop in accuracy. This pattern held consistently across two independently constructed dataset baselines. These findings suggest that, for clinical screening tasks where questionnaire data is often incomplete, Bayesian networks offer a meaningful robustness advantage that is not automatically matched by machine learning models commonly assumed to handle missing data natively.

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